

Project 2: Thermal-Hydraulic Design and Metallurgical Evaluation of a Seawater Steam Surface Condenser

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Abstract

In coastal desalination infrastructure and coastal thermal power generating stations, seawater-cooled surface condensers represent critical thermal nodes governing overall plant efficiency and lifecycle generation costs. While copper-based alloys such as Aluminum-Brass have historically dominated seawater tubing applications due to high intrinsic thermal conductivity and biofouling resistance, their susceptibility to erosion-corrosion, sulfide attack, and progressive corrosion-product scaling poses substantial operational burdens.

This project delivers a comprehensive design and comparative metallurgical evaluation for a 24.0 MW steam condenser using **HTRI Xist (TEMA AES)**. A baseline Aluminum-Brass configuration (1.245 mm wall) is systematically contrasted against Titanium Grade 2 in both standard (1.245 mm) and thin-wall (0.559 mm) profiles. Furthermore, conventional TEMA fouling standards are critically reviewed against Heat Exchange Institute (HEI) standards. Finally, an analytical thermal resistance model formulated in MATLAB is evaluated against HTRI results, unveiling a 10.3% underestimation driven by internal tubeside velocity attenuation and convective degradation ($h_i \propto V^{0.8}$).

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1 Introduction and Theoretical Metallurgy Framework

The core task involves the thermal-hydraulic design of a surface condenser condensing 40,198 kg/h of exhaust steam at 154.2°C (500 kPa) using raw seawater entering at 18.0°C and exiting at 42.5°C.

1.1 Degradation and Corrosion Mechanisms: Copper Alloys vs. Titanium

- **Copper-Based Alloys (Aluminum-Brass, Cu-Ni):** Copper alloys rely entirely on a delicate cuprous oxide (Cu_2O) passivating film. Under dynamic seawater turbulence exceeding a critical threshold ($\approx 2.0\text{--}3.0\text{ m/s}$), this protective film shears off, triggering accelerated *erosion-corrosion*. Furthermore, in polluted coastal seawater containing dissolved sulfides, copper oxides convert into non-adherent copper sulfides, inducing catastrophic localized pitting.
- **Titanium Grade 2:** Titanium instantly oxidizes in aqueous media to form a tenacious, self-healing rutile/anatase titanium dioxide (TiO_2) passive layer. This ceramic-grade film resists dynamic fluid velocities beyond 20 m/s, granting complete immunity to pitting, crevice corrosion, and erosion-corrosion in raw seawater across industrial temperature operating windows.

1.2 Fouling Dynamics and Resistance Mitigation

- **Aluminum-Brass:** Cuprous ions (Cu^{2+}) exert a biocidal effect that retards macro-biological attachment (e.g., barnacles, algae). However, progressive substrate corrosion forms heavy, porous scale composed of basic copper salts and carbonates, which drastically impedes conductive heat transport.
- **Titanium:** Titanium lacks biocidal surface activity. However, it exhibits a non-polar, low surface energy state that prevents scale adhesion. Moreover, the superior mechanical integrity of titanium allows continuous high-velocity flow and the deployment of continuous mechanical sponge-ball cleaning systems (e.g., Taprogge systems), sustaining an active *self-cleaning* shear effect.

1.3 Critique of Standard TEMA Seawater Fouling Tables

Standard TEMA tables tabulate seawater fouling factors solely as functions of bulk temperature and velocity, ignoring tube metallurgy. This is flawed for seawater applications:

1. **Corrosion Products as Internal Fouling:** On copper alloy tubing, the tube material actively corrodes, generating internal scale resistance (R_f). Conversely, titanium exhibits near-zero metal loss ($< 0.001\text{ mm/year}$), completely eliminating the corrosion fouling component.
2. **Surface Energy and Roughness:** Titanium retains a smooth, glass-like surface, suppressing particulate sedimentation, whereas copper alloys develop high surface roughness that promotes scale nucleation.

3. **Velocity Limits:** TEMA bases its default figures on velocities of ≤ 1.5 m/s. Titanium permits operational velocities of 3–4 m/s, drastically lowering fouling deposition rates.

Consequently, specialized condenser standards (e.g., **HEI Standards for Steam Surface Condensers**) recommend reducing design fouling resistances by roughly 50% when replacing copper alloys with titanium ($0.00040 \rightarrow 0.00018\text{--}0.00020$ m²K/W).

2 Exchanger Architecture and Parameter Selection

- **TEMA Type AES Configuration:**

- *Front Head (Type A):* Removable bonnet cover allowing straight access for tubeside mechanical rod cleaning without dismantling seawater piping headers.
- *Shell (Type E):* Standard single-pass crossflow condensing shell.
- *Rear Head (Type S - Floating Head with Backing Device):* Accommodates severe differential thermal expansion between carbon steel shell and tubes subjected to high terminal thermal gradients ($\Delta T = 154.2^\circ\text{C} - 18.0^\circ\text{C} = 136.2^\circ\text{C}$).

- **Bundle Geometry:** 30° Triangular pitch provides maximal packaging density for clean condensing steam.
- **Baffling Scheme:** No-Tubes-In-Window (NTIW) segmental baffles with 25% diametral cut and 700 mm central spacing were implemented to suppress acoustic resonance and crossflow tube bundle vibration while keeping shell pressure drop well beneath the 25.0 kPa ceiling.

3 Thermal-Hydraulic Design & Metallurgical Evaluation

3.1 Material Comparison: Base Aluminum-Brass vs. Standard Titanium

The base exchanger was sized in HTRI **Design Mode** using Aluminum-Brass ($k \approx 100$ W/m · K, $t_w = 1.245$ mm). Subsequently, the shell envelope ($D_{\text{shell}} = 1168.4$ mm) was locked in **Rating Mode** to evaluate standard Titanium Grade 2 ($k \approx 21.9$ W/m · K, $t_w = 1.245$ mm).

Table 1: Performance comparison: Aluminum-Brass vs. Standard Titanium (Fixed Shell Diameter).

Design Parameter	Aluminum-Brass	Titanium (Grade 2)	Engineering Variation
Execution Mode	Design Mode	Rating Mode	Fixed Envelope
Shell Inside Diameter	1168.4 mm	1168.4 mm	Constant
Tube Metallurgy	Al-Brass (78Cu, 2Al)	Titanium Grade 2	Lower Conductivity
Tube Wall Thickness	1.245 mm	1.245 mm	Constant
Total Tube Count (N_t)	943	1184	+25.5% (+241 tubes)
Number of Tube Passes	2 passes	1 pass	Reconfigured
Overall Coefficient (U)	1531.2 W/m ² K	1210.3 W/m ² K	−21.0%
Tube Metal Resistance	1.97%	7.99%	$\approx 4\times$ increase
Effective Area (A)	131.68 m ²	166.23 m ²	+26.2%
Tubeside Pressure Drop	30.77 kPa	5.16 kPa	−83.2%
Shellside Pressure Drop	10.10 kPa	9.61 kPa	Constant (< 25 kPa)
Overdesign Margin	0.92%	0.90%	Target Matched

Engineering Discussion: Due to titanium’s low thermal conductivity (21.9 vs. 100 W/m·K), wall conductive resistance quadruples (1.97% $\rightarrow$ 7.99%). Delivering the 24.0 MW duty within identical shell dimensions requires adding 241 tubes, demonstrating that simple 1:1 metallurgy replacement imposes a severe capital penalty.

HTRI		HEAT EXCHANGER SPECIFICATION SHEET				Page 1 SI Units	
Customer				Job No.			
Address				Reference No.			
Plant Location				Proposal No.			
Service of Unit				Date 2/22/2026 Rev			
Size 1168.4 x 2438 mm Type AES Horizontal				Item No.			
Surf/Unit (Gross/Eff) 137.59 / 131.68 m2 Shell/Unit 1				Connected In 1 Parallel 1 Series			
				Surf/Shell (Gross/Eff) 137.59 / 131.68 m2			
PERFORMANCE OF ONE UNIT							
Fluid Allocation		Shell Side			Tube Side		
Fluid Name		steam			sea water		
Fluid Quantity, Total kg/hr		40198			820798		
Vapor (In/Out)		40198					
Liquid		40198			820798		
Steam		40198			820798		
Water		40198			820798		
Noncondensables							
Temperature (In/Out) C		154.20			18.00		
Specific Gravity					1.0132		
Viscosity mN-s/m2		0.0141			1.0524		
Molecular Weight, Vapor							
Molecular Weight, Noncondensables							
Specific Heat kJ/kg-C		1.9769			4.3145		
Thermal Conductivity W/m-C		0.0304			0.6002		
Latent Heat kJ/kg		2130.2			2132.6		
Inlet Pressure kPa		500.00			400.00		
Velocity m/s		8.75			2.19		
Pressure Drop, Allow/Calc kPa		25.000			10.097		
Fouling Resistance (min) m2-K/W		0.000000			0.000400		
Heat Exchanged 24002421 W					MTD (Corrected) 120.1 C		
Transfer Rate, Service 1517.3 W/m2-K		Clean 5182.8 W/m2-K			Actual 1531.2 W/m2-K		
CONSTRUCTION OF ONE SHELL				Sketch (Bundle/Nozzle Orientation)			
		Shell Side		Tube Side			
Design/Test Pressure kPaG		517.11 /		517.11 /			
Design Temperature C		182.22		71.11			
No Passes per Shell		1		2			
Corrosion Allowance mm		3.175		3.175			
Connections		In mm 1 @ 300.00		1 @ 387.35			
Size & Rating		Out mm 1 @ 205.00		1 @ 387.35			
Intermediate		@		@			
Tube No. 943		OD 19.050 mm		Thk(Avg) 1.245 mm		Length 2.438 m	
Tube Type Plain				Material Aluminum brass (78 Cu, 2 Al)		Pitch 25.337 mm	
Shell Carbon steel		ID 1168.4 OD 1193.8 mm		Shell Cover		Tube pattern 30	
Channel or Bonnet				Channel Cover		(Remove.)	
Tubesheet-Stationary				Tubesheet-Floating			
Floating Head Cover				Impingement Plate		Rods	
Baffles-Cross		Type NTIW-Seg.		%Cut (Diam) 25		Spacing(c/c) 380.67	
Baffles-Long				Seal Type None		Inlet 785.95 mm	
Supports-Tube				U-Bend		Type None	
Bypass Seal Arrangement 5 pairs seal strips				Tube-Tubesheet Joint Expanded (No groove)			
Expansion Joint				Type			
Rho-V2-Inlet Nozzle 9547.0 kg/m-s2				Bundle Entrance 1280.7		Bundle Exit 18.70 kg/m-s2	
Gaskets-Shell Side Mach. Mtl. (Kammprofile\Flex. Face)				Tube Side Mach. Mtl. (Kammprofile\Flex. Face)			
- Floating Head Mach. Mtl. (Kammprofile\Flex. Face)							
Code Requirements				TEMA Class R			
Weight/Shell 6885.4 kg		Filled with Water 11489 kg		Bundle 3002.8 kg			
Remarks: Use deresonating baffles.							

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Figure 1: HTRI TEMA Specification Sheet – Base Aluminum-Brass Design ($t_w = 1.245$ mm, 943 Tubes).

HTRI		HEAT EXCHANGER SPECIFICATION SHEET				Page 1 SI Units	
Customer				Job No.			
Address				Reference No.			
Plant Location				Proposal No.			
Service of Unit				Date 2/23/2026		Rev	
Size 1168.4 x 2438 mm Type AES Horizontal				Item No.		Connected In 1 Parallel 1 Series	
Surf/Unit (Gross/Eff) 172.75 / 166.23 m ² Shell/Unit 1				Surf/Shell (Gross/Eff) 172.75 / 166.23 m ²			
PERFORMANCE OF ONE UNIT							
Fluid Allocation		Shell Side			Tube Side		
Fluid Name		steam			sea water		
Fluid Quantity, Total kg/hr		40198			820798		
Vapor (In/Out)		40198					
Liquid		40198			820798		
Steam		40198			820798		
Water		40198			820798		
Noncondensables							
Temperature (In/Out) C		154.20			18.00		
Specific Gravity		0.9031			1.0132		
Viscosity mN-s/m ²		0.0141			1.0524		
Molecular Weight, Vapor							
Molecular Weight, Noncondensables							
Specific Heat kJ/kg-C		1.9769			4.3145		
Thermal Conductivity W/m-C		0.0304			0.6002		
Latent Heat kJ/kg		2130.2			2132.4		
Inlet Pressure kPa		500.00			400.00		
Velocity m/s		7.72			0.89		
Pressure Drop, Allow/Calc kPa		25.000			50.000		
Fouling Resistance (min) m ² -K/W		0.000000			0.000400		
Heat Exchanged 24001169 W		MTD (Corrected) 120.4 C					
Transfer Rate, Service 1199.6 W/m ² -K		Clean 2731.5 W/m ² -K			Actual 1210.3 W/m ² -K		
CONSTRUCTION OF ONE SHELL				Sketch (Bundle/Nozzle Orientation)			
Design/Test Pressure kPaG		Shell Side 517.11 /		Tube Side 517.11 /			
Design Temperature C		182.22		71.11			
No Passes per Shell		1		1			
Corrosion Allowance mm		3.175		3.175			
Connections		In mm 1 @ 300.00		1 @ 387.35			
Size & Rating		Out mm 1 @ 205.00		1 @ 387.35			
Rating		Intermediate @		@			
Tube No. 1184		OD 19.050 mm		Thk(Avg) 1.245 mm		Length 2.438 m	
Tube Type Plain		Material Titanium-grade 2		Pitch 25.337 mm		Tube pattern 30	
Shell Carbon steel		ID 1168.4 OD 1193.8 mm		Shell Cover (Remove.)			
Channel or Bonnet		Channel Cover					
Tubesheet-Stationary		Tubesheet-Floating					
Floating Head Cover		Impingement Plate Rods					
Baffles-Cross		Type NTIW-Seg.		%Cut (Diam) 25		Spacing(c/c) 700.00	
Baffles-Long		Seal Type None					
Supports-Tube		U-Bend		Type None			
Bypass Seal Arrangement 5 pairs seal strips		Tube-Tubesheet Joint		Expanded (No groove)			
Expansion Joint		Type					
Rho-V2-Inlet Nozzle 9547.0 kg/m-s ²		Bundle Entrance 1245.9		Bundle Exit 17.37		kg/m-s ²	
Gaskets-Shell Side Mach. Mtl. (Kammprofile\Flex. Face)		Tube Side Mach. Mtl. (Kammprofile\Flex. Face)					
- Floating Head Mach. Mtl. (Kammprofile\Flex. Face)							
Code Requirements		TEMA Class R					
Weight/Shell 5515.3 kg		Filled with Water 9969.8 kg		Bundle 2073.7 kg			
Remarks: Use deresonating baffles.							
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Figure 2: HTRI TEMA Specification Sheet – Standard-Gauge Titanium Design ($t_w = 1.245$ mm, 1184 Tubes).

3.2 Optimization via Thin-Wall Titanium ($t_w = 0.559$ mm)

Titanium’s high tensile and fatigue strength permits utilizing thin-wall tubing (0.559 mm $\approx$ 500 μ m) without risking hydraulic collapse.

Table 2: Optimization impact: Standard Titanium vs. Thin-Wall Titanium.

Parameter	Standard Titanium	Thin-Wall Titanium	Net Benefit
Tube Thickness (t_w)	1.245 mm	0.559 mm	−55.1% reduction
Total Tube Count (N_t)	1184	1076	108 fewer tubes
Effective Surface Area	166.23 m ²	150.86 m ²	−9.2% reduction
Overall Coefficient (U)	1210.3 W/m ² K	1334.3 W/m ² K	+10.2% enhancement
Metal Wall Resistance	7.99%	3.81%	Halved bottleneck
Tubeside Pressure Drop	5.16 kPa	4.76 kPa	−7.7%
Tubeside Bulk Velocity	0.89 m/s	0.84 m/s	Safe low erosion

Quantitative Comparison vs. Base Al-Brass:

$$\Delta N = N_{\text{Thin-Ti}} - N_{\text{Al-Brass}} = 1076 - 943 = +133 \text{ tubes} \quad (+14.1\%) \quad (1)$$

Thinning the tube wall recovers more than half of the area penalty seen in standard-gauge titanium.

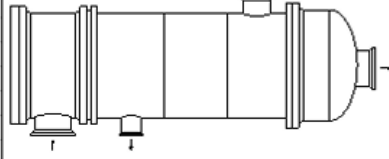
HTRI		HEAT EXCHANGER SPECIFICATION SHEET										Page 1 SI Units																																					
Customer												Job No.																																					
Address												Reference No.																																					
Plant Location												Proposal No.																																					
Service of Unit												Date		2/24/2026		Rev																																	
Size												1168.4 x 2438		mm		Type		AES		Horizontal		Connected In		1		Parallel		1		Series																			
Surf/Unit (Gross/Eff)												157 / 150.86		m2		Shell/Unit		1		Surf/Shell (Gross/Eff)		157 / 150.86		m2																									
PERFORMANCE OF ONE UNIT																																																	
Fluid Allocation								Shell Side										Tube Side																															
Fluid Name								steam										sea water																															
Fluid Quantity, Total								kg/hr		40198										820798																													
Vapor (In/Out)								40198																																									
Liquid																		40198										820798										820798											
Steam								40198																				820798										820798											
Water																		40198										820798										820798											
Noncondensables																																																	
Temperature (In/Out)								C		154.20										151.12										18.00										42.50									
Specific Gravity																		0.9031										1.0132										0.9946											
Viscosity								mN-s/m2		0.0141										0.1797										1.0524										0.6218									
Molecular Weight, Vapor																																																	
Molecular Weight, Noncondensables																																																	
Specific Heat								kJ/kg-C		1.9769										4.5761										4.3145										4.3165									
Thermal Conductivity								W/m-C		0.0304										0.6864										0.6002										0.6346									
Latent Heat								kJ/kg		2130.2										2132.4																													
Inlet Pressure								kPa												500.00										400.00																			
Velocity								m/s												7.70										0.84																			
Pressure Drop, Allow/Calc								kPa		25.000										9.612										50.000										4.758									
Fouling Resistance (min)								m2-K/W												0.000000										0.000400																			
Heat Exchanged								24001169 W										MTD (Corrected)										120.4 C																					
Transfer Rate, Service								1321.9 W/m2-K										Clean 3081.3 W/m2-K										Actual 1334.3 W/m2-K																					
CONSTRUCTION OF ONE SHELL																																																	
								Shell Side										Tube Side										Sketch (Bundle/Nozzle Orientation)																					
Design/Test Pressure								kPaG		517.11 /										517.11 /																													
Design Temperature								C		182.22										71.11																													
No Passes per Shell								1										1																															
Corrosion Allowance								mm		3.175										3.175																													
Connections		In		mm		1 @		300.00						1 @		387.35																																	
		Out		mm		1 @		205.00						1 @		387.35																																	
Rating		Intermediate		@		@						@		@																																			
Tube No.								1076		OD		19.050 mm		Thk(Avg)		0.559 mm		Length		2.438 m		Pitch		25.337 mm																									
Tube Type								Plain		Material										Titanium-grade 2										Tube pattern		30																	
Shell								Carbon steel		ID		1168.4		OD		1193.8 mm		Shell Cover										(Remove.)																					
Channel or Bonnet																		Channel Cover																															
Tubesheet-Stationary																		Tubesheet-Floating																															
Floating Head Cover																		Impingement Plate										Rods																					
Baffles-Cross								Type										NTIW-Seg.		%Cut (Diam)		25		Spacing(c/c)		700.00		Inlet		821.38 mm																			
Baffles-Long																		Seal Type										None																					
Supports-Tube																		U-Bend												Type		None																	
Bypass Seal Arrangement								5		pairs seal strips		Tube-Tubesheet Joint										Expanded (No groove)																											
Expansion Joint																		Type																															
Rho-V2-Inlet Nozzle								9547.0		kg/m-s2		Bundle Entrance										1250.8		Bundle Exit		17.44		kg/m-s2																					
Gaskets-Shell Side								Mach. Mtl. (Kammprofile\Flex. Face)										Tube Side										Mach. Mtl. (Kammprofile\Flex. Face)																					
- Floating Head								Mach. Mtl. (Kammprofile\Flex. Face)																																									
Code Requirements																		TEMA Class										R																					
Weight/Shell								5141.0		kg		Filled with Water										9680.2		kg		Bundle										1699.8		kg											
Remarks: Use deresonating baffles.																																																	
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Figure 3: HTRI TEMA Specification Sheet – Optimized Thin-Wall Titanium Design ($t_w = 0.559$ mm, 1076 Tubes).

4 HTRI Runtime Diagnostics and Vibration Analysis

During execution, four specific warning messages were diagnosed:

1. **Acoustic Vibration Warning:** HTRI flagged potential fundamental transverse acoustic resonance driven by high-velocity vapor vortex shedding. *Resolution:* HTRI recommended installing deresonating baffles at $0.45 \times D_{\text{shell}}$. This detunes the acoustic cavity natural frequency without impeding crossflow condensation.
2. **Thermal Conductivity Temperature Limit (Al-Brass):** Bulk condensing steam temperatures (154.2°C) exceeded the experimental database validation cap of 176.1°F (80°C) for material code 45. Thermal conductivity was clamped at the bound; resulting deviation in U was negligible ($< 0.2\%$).
3. **Rear Head Internal Diameter Limitation:** A geometric check on the Type S floating head indicated tight clearance between the floating tubesheet backing device and shell ID. This is addressed during structural mechanical detailing.
4. **ASME Section VIII Div. 1 Disclaimer:** Standard HTRI notice declaring vessel wall thicknesses and weights are preliminary estimations and require formal PVELite mechanical verification.

	Runtime Messages Page 1 Released to the following HTRI Member Company: 989144785264
	Xist 7.3.2 2/22/2026 16:33 SN: 46639- SI Units
Design - Horizontal Multipass Flow TEMA AES Shell With NTIW-Segmental Baffles	
Unit ID 100 - WARNING MESSAGES (CALCULATIONS CONTINUE)	
The maximum process temperature is outside the correlational range of the thermal conductivity correlation. The thermal conductivity was calculated at the temperature limit. For tube metal number 45 this temperature limit is 176.1°F.	
Deresonating baffles placed at $0.45 \times \text{Shell ID}$ will resolve the first fundamental transverse mode for acoustic vibration. Two deresonating baffles placed at $0.18 \times \text{Shell ID}$ and $0.45 \times \text{Shell ID}$ will resolve up to 5 transverse modes for acoustic vibration. User must ensure that the proper number of tubes are removed for placement of deresonating baffles. Xist HAS NOT taken this reduction in heat transfer surface area into account.	
ASME Section VIII Div. 1 Code was used to ESTIMATE all pressure vessel dimensions and weights. The dimensions in this report cannot be used to fabricate the vessel.	
Rear head ID may be too large to permit a floating tubesheet smaller than the shell.	

Figure 4: HTRI Xist Runtime Messages and Acoustic Vibration Diagnostics.

5 Analytical Validation via MATLAB and Root-Cause Analysis

An in-house 1D thermal resistance script was deployed to predict the required tube count when transitioning from Al-Brass to thin-wall Titanium:

$$\frac{1}{U_o} = R_{\text{wall}} + R_{\text{other}} = \frac{D_o \ln(D_o/D_i)}{2k} + \frac{D_o}{D_i} R_{\text{conv,in}} + R_{\text{conv,out}} + R_{\text{foul}} \quad (2)$$

Assuming R_{other} remains invariant and holding thermal duty constant:

$$N_{\text{est}} = N_{\text{base}} \times \left(\frac{U_{\text{base}}}{U_{\text{new}}} \right) \quad (3)$$

Table 3: MATLAB 1D analytical model vs. HTRI multi-segment rating.

Metric	Al-Brass (Base)	Titanium (MATLAB)	Titanium (HTRI Actual)
Overall Coefficient (U)	1531.20 W/m ² K	1497.49 W/m ² K	1334.30 W/m ² K
Required Tube Count (N_t)	943	965	1076
Model Error Relative to HTRI	—	−10.32%	Benchmark

U Base (Al-Brass): 1531.20 U New (Titanium): 1497.49
 Tube Count (Base): 943 Tube Count (Est): 965 Tube Count (HTRI): 1076
 Error: 10.32%

Figure 5: MATLAB Command Window Output: Analytical Sizing Discrepancy (Error = 10.32%).

5.1 Root-Cause Diagnosis of the 10.3% Underestimation

The MATLAB script underestimated tube count by 111 tubes due to coupled **thermal-hydraulic feedback**:

1. **Internal Diameter Expansion:** Thinning the tube wall from 1.245 mm to 0.559 mm under a fixed OD (19.05 mm) expanded the inner diameter:

$$D_{i,\text{Al-Brass}} = 16.56 \text{ mm} \quad \longrightarrow \quad D_{i,\text{Thin-Ti}} = 17.93 \text{ mm} \quad (4)$$

2. **Velocity Attenuation:** Flow cross-sectional area per tube expanded by 17.3%. With total mass throughput held constant, seawater bulk velocity dropped:

$$V = \frac{\dot{m}}{\rho A_{\text{flow}}} \implies V_{\text{Std-Ti}} = 0.89 \text{ m/s} \quad \longrightarrow \quad V_{\text{Thin-Ti}} = 0.84 \text{ m/s} \quad (5)$$

3. **Convective Heat Transfer Degradation:** Seawater forced convection obeys the Dittus-Boelter / Sieder-Tate correlation:

$$h_i \propto \frac{k}{D_i} Re^{0.8} \propto \frac{V^{0.8}}{D_i^{0.2}} \quad (6)$$

The drop in velocity combined with the expanded diameter degraded h_i .

Because the MATLAB script assumed R_{other} was constant, it overestimated U (1497.5 vs. 1334.3 W/m²K) yielding an unconservative tube count estimate. This demonstrates that tube gauge modifications cannot be treated as isolated conductive perturbations.

6 Summary and Conclusions

1. Substituting Aluminum-Brass with standard-gauge Titanium Grade 2 imposes a 25.5% tube count penalty ($943 \rightarrow 1184$ tubes) due to titanium's lower conductivity ($21.9 \text{ W/m} \cdot \text{K}$).
2. Sizing with thin-wall titanium (0.559 mm) recovers 45% of this surface penalty, requiring 1076 tubes (+14.1% over base), cutting metal resistance to 3.81% while yielding complete immunity to seawater erosion-corrosion.
3. Coupling 1D conduction with internal hydraulics is critical; internal diameter enlargement attenuates velocity, degrading internal convection ($h_i \propto V^{0.8}$) by 10.3%.

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